

Document-level relation extraction with hierarchical dependency tree and bridge path

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ABSTRACT

The inter-sentence relation in a document is characterized by complex contextual information, large span of correlation and many kinds of relations, leading to the poor effect of sentence-level relation extraction models when addressing inter-sentence relations. Graph networks have been widely used in the research of document-level relation extraction due to their advantages in modeling local structural features and long-distance context dependencies. However, most previous studies modeled document in a coarse-grained manner, which ignores the richness and otherness of hierarchical features in a document. Consequently, contextual information modeling is not sufficient and fails to participate in deep reasoning efficiently. In this paper, we propose a document-level relation extraction model based on the Hierarchical Dependency Tree and Bridge Path (HDT-BP). The model uses sentence as a unit to independently extract the fine-grained features of each hierarchy and reconstructs the chain-structured document based on multiple dependent relationships into a hierarchical dependency tree. Moreover, the relational bridge entity is introduced during relation extraction to improve the model performance by modeling the bridge path feature. Experimental results demonstrate that our model exhibits superior performance on the DocRED dataset and achieves a significant improvement in extracting relational facts that never appeared in the training set. Extensive additional experiments further verify the effectiveness of our model.

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1. Introduction

Relation extraction (RE) is a crucial field of research in information extraction (IE) that focuses on extracting semantic relationships from among all entities in unstructured sequential text. RE plays an essential role in constructing complex systems, including knowledge graphs, search engines, and question-answering systems [1,2]. Earlier studies on RE concentrated mainly on extracting relations between entities within a sentence [3,4], and various advanced models based on deep neural networks (DNN) have been proposed [5–9]. Nonetheless, relations between entities in real-life situations usually cross multiple sentences and present complicated contextual dependencies, which make it hard for the sentence-level RE (SenRE) models

to capture the complex and diverse inter-sentence relations in a document [10].

Fig. 1 shows an annotated sample of document-level RE (DocRE), which contains six sentences, five entities, and three relations. The first relation is an intra-sentence relation template that denotes 'Demerara' (location entity) as being in the administrative area of 'Demerara-Mahaica' (location entity). It can be extracted by requiring the model to possess the single-sentence modeling ability. The second relation is an inter-sentence relation template that indicates 'Emerson Samuels' (person entity) was born in 'Nabacalis' (location entity). This relation spans five sentences with the head entity in sentence (1) and the tail entity in sentence (5). The supporting information for this relation comes from sentences (3) and (4): the word 'here' in sentence (5) refers to the word 'hospital' in sentence (4), and the word 'community' in sentence (3) refers to the head entity 'Nabacalis' in sentence (1). The complete relation support chain (RSC) is 'Emerson Samuels'-'born'-'here'-'hospital'-'community'-'Nabacalis'. The third relation implies that 'Rudolph Dunbar' (person entity) was born in 'Nabacalis' (location entity), and this pattern is analogous to the second relation but with an extended scope. By observing

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(1) Nabacalis is a <u>community</u> on the <u>East Coast</u> of Demerara , in the Demerara - Mahaica <u>region</u> .
(2)
(3) The <u>community</u> is served by the Dr. CC Nicholson <u>Hospital</u>
(4) The <u>hospital</u> was established as a medical center in 1994 with funds from the London
(5) The artist Emerson Samuels was <u>born</u> and brought up <u>here</u> , studying at the Golden Grove Methodist School .
(6) Also <u>born here</u> was conductor Rudolph Dunbar .
Head Entity: (1) Demerara (LOC)
Tail Entity: (1) Demerara – Mahaica (LOC)
Relation: located in the administrative territorial entity
Head Entity: (5) Emerson Samuels (PER)
Tail Entity: (1) Nabacalis (LOC)
Relation: place of birth
Head Entity: (6) Rudolph Dunbar (PER)
Tail Entity: (1) Nabacalis (LOC)
Relation: place of birth

Fig. 1. Sample from the DocRED [10] dataset. The green text represents location entities, the blue text represents person entities, and the underlined text represents the important contextual information in the relation support chain.

the sample in Fig. 1, it is evident that DocRE encounters various challenges, including intricate contexts, multiple mentions of entity references, numerous discrete RCS, and a broad range of entity types.

In response to the aforementioned challenges, earlier designs of DocRE models employed convolutional neural networks (CNN) or recurrent neural networks (RNN) as the primary encoder [11–14]. Such models assume that entities in a document are mentioned at most twice [13,15] and independently design two models for intra-sentence and inter-sentence RE [12,16,17]. The former loses some features regarding entity mentions, while the latter raises the overall parameter size and training cost. Recently, inspired by the benefits of structure-dependent information between entities for enhancing the SenRE performance [18], graph neural network (GNN)-based research on DocRE algorithms has become increasingly attractive [19,20]. Furthermore, GNNs can model non-local dependent items with ambiguous surface forms [18]. Typically, these methods utilize a general word embedding encoder (like long short-term memory (LSTM) or pre-trained language model (PLM)) to construct the feature representation of the document. Subsequently, the structural information between entities is introduced by constructing a well-designed heterogeneous graph in which the feature representation is updated accordingly with the aggregation completion of the GNN. Finally, the classifier on top of the model utilizes entity pair features to complete DocRE.

However, analysis of recent studies has identified three limitations of current GNN-based models: (i) Previous works tend to perform one-shot modeling of the entire document in a coarse-grained manner, causing a loss of local features and the sentence hierarchy concept. In particular, when using the PLM as the embedding layer, such as BERT [21], the part of the word sequence that exceeds a length of 512 may be cropped, resulting in the

loss of important information. (ii) Excessive attention given to the semantics and structure of entities marginalizes the context during the initial word embedding stage. Since there is no in-depth modeling, the contextual information of entities is less utilized during the model reasoning process. However, we intuitively believe that an entity and its context are highly correlated, and in most cases, the contextual information can reflect some of the features of entity. (iii) Observation suggests that the most important information that is conducive to RE has a co-occurrence relation with the head/tail entity, yet previous studies have not discriminated the modeling weight of the contextual information between different local regions.

To address the aforementioned issues, this paper proposes a DocRE model based on the Hierarchical Dependency Tree and Bridge Path (HDT-BP). As shown in Fig. 2, the model first employs a novel document reconstruction method that implements fine-grained modeling and structural reorganization of the chained structure document sequence on a sentence-by-sentence basis, representing the document as a tree diagram with explicit hierarchical dependencies. In this stage, the model builds an embedding layer of input tokens and generates the feature representation of nodes at each hierarchy based on intermediate operators such as pooling and LSTM. The final tree diagram contains node features at four hierarchies: document, sentence, context, and the mention of the entity. Second, by introducing co-occurrence and co-reference relations between nodes of different hierarchies, the model realizes full information depth modeling with different modeling weights in different regions. In this stage, three types of edges (document-sentence, sentence-mention, mention-context) are added between nodes of different hierarchies to introduce structural dependence relationships, completing the construction of the hierarchical dependency tree (HDT). Graph convolutional network (GCN) is used to aggregate and update the features

of HDT through multiple rounds, and it extracts the updated mention/document nodes to construct an entity correlation graph (ECG) for modeling the relation features between entities. Finally, in addition to using the node features of the entity pair and document, the model introduces relational bridge entities (third-party entities that have co-occurrence relation with both the head and tail entity) to model the bridge path features to enhance the classification features, improving the prediction accuracy of DocRE.

The contributions of this paper can be summarized as follows:

- A DocRE model based on the hierarchical dependency tree and bridge path is proposed. This model can perform fine-grained joint modeling for information in different levels or regions in a document and realize simultaneous extraction of inter-sentence and intra-sentence relations.
- We propose a fine-grained document reconstruction method to reconstruct the chain structure into a tree structure by capturing hierarchical structural dependency information. This method assigns independent weights to the features of different regions, improving the utilization efficiency of features at each hierarchy.
- A relation modeling method based on bridge path information enhancement is designed. By introducing a bridge entity between entity pairs, an implicit dependency structure is constructed in the document, and it effectively improves the model performance.
- Our model exhibits superior performance on the DocRED dataset and achieves a significant improvement in terms of extracting relation facts that have never appeared in the training set.

2. Related work

RE plays an important role in the field of IE and is a key technology for building complex intelligent systems such as knowledge graphs or recommendation systems [10]. At present, sentence-level RE has achieved remarkable results [22–25], but relational dependencies in real contexts often exist in the form of cross-sentences in a document. Benefitting from the rise of DNN technology, a large number of models for DocRE based on deep learning have been proposed [18–20,26,27]. These models can be roughly classified into three types: sequence-based models, attention-based models and graph-based models.

The sequence-based model designed with the CNN or RNN as the core encoding layer is the earliest method applied to DocRE, and it needs to be combined with feature engineering to extract document features. Drawing on the experience of sentence-level RE, the dependencies between entities in a document are also introduced by the model during the preprocessing stage. Specifically, Zhou et al. [11] proposed a novel syntactic-semantic information mining method for DocRE that consists of a feature-based model, a tree-kernel-based model and an LSTM layer. Gu et al. [17] built a machine learning system that models intra-sentence and inter-sentence relations separately and uses simple and effective text features to extract relations in a document based on the maximum entropy model. On this basis, Gu et al. [12] further introduced the CNN into the model to perform different levels of training between intra-sentence and inter-sentence. Nguyen and Verspoor [13] incorporated character feature-based word representations into a standard CNN-based RE model and tried to learn word feature representations from character features using the CNN or LSTM, respectively. Sequence-based models have achieved certain results in DocRE. However, the limitations of model structure make it difficult to capture long-distance context-dependent information, and these

approaches cannot model the structural-dependent features of the inter-sentence entity in a document.

Research on sentence-level RE has demonstrated that deep modeling of structural dependencies between entities can improve model performance [18]. The main idea of attention-based models is to perform comprehensive and deep modeling of the structural dependency information between entities within a document. Specifically, Verga et al. [15] first introduced multi-instance learning [28] into DocRE. The method uses a self-attention mechanism to align the entire document and make pairwise predictions. All pairwise mention scores allow the model to aggregate through an adapted multi-instance learning approach to form feature representations for entity pairs. Huang et al. [29] considered that the PLM gives a more balanced attention weight to all words when executing RE task. This phenomenon limits the ability to model important words in a document and inhibits the performance of the PLM. Based on this idea, Huang et al. [29] designed an E2GRE architecture to encourage the PLM to focus on entities and supporting sentences in a document. Zhou et al. [30] proposed adaptive thresholding and localized context pooling to solve the multi-label and multi-entity problems in DocRE. Xu et al. [31] designed an SSAN based on Transformer, which incorporates the structure of coreference dependency and co-occurrence dependency between entities into a standard self-attention layer and employs that structure throughout the entire encoding stage.

Recently, the design of DocRE algorithms based on the GNN has become the mainstream trend. It has been proven that its graph structure has significant advantages when encoding long-distance dependencies and inter-sentence information [26,32,33]. Specifically, Guo et al. [34] used the syntactic dependency tree of the entire document as the model input and utilized the attention mechanism to learn the weights of all edges in the graph with an end-to-end mode. This method selectively focuses on local structures that are valuable for RE and implements automatic pruning. Christopoulou et al. [20] proposed an edge-oriented GNN. This network utilizes different types of nodes and edges to construct relational graphs, and the inference mechanism of graph edges uses multi-instance learning to capture the inter-sentence and intra-sentence dependencies of entities. Nan et al. [18] proposed a GNN based on structured attention units. This network is able to generate specific dependency structures for different tasks and incrementally aggregate relevant information to capture non-local interactions between entities. Tran et al. [35] considered that the previous GNN based on directed edges cannot effectively model the features of nodes and edges. The authors suggested explicit computation of nodes in graph models and designed two novel regularization mechanisms to improve feature representation in DocRE. Xu et al. [36] introduced graph reconstruction methods commonly used in the field of neural machine translation. This method reconstructs the document path to guide the model to focus more attention on entity pairs with relation, which further improves the model performance.

There are some related works outside the conventional methodology that are relevant to the background of this study. In terms of relation annotations, Nasar et al. [37] noted that the major design challenges of RE include the selection of annotation schemes and inferencing algorithms. Different annotation schemes have a profound influence on the learning effect of deep models. Wang et al. [38] proposed a novel handshaking annotations scheme that aligns the boundary tokens of entity pairs under each relation type to mitigate the problem of exposure bias in joint learning, and Tan et al. [39] argued that there is a discrepancy between existing causal relation annotation guidelines and the linguistic theory-oriented traditional causal relation corpora. They crafted an annotation guideline that aligns with both existing causal relations and linguistics. Van et al. [40] considered

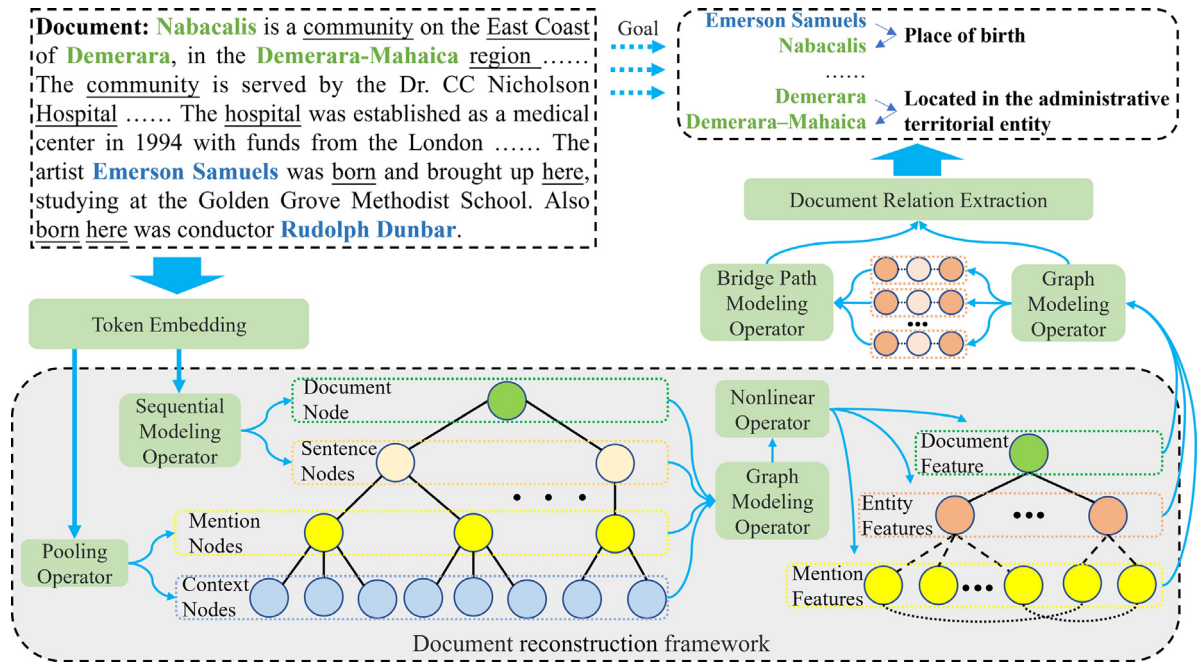


Fig. 2. Overall framework of the proposed model. According to the novel reconstruction method design, the chain structure of the document is converted into a tree diagram that explicitly expresses the structural dependency relations. Based on various operators in the framework, the model gradually generates the node representations of each hierarchical feature and updates the HDT by aggregating the neighborhood dependency feature. Co-occurrence and co-reference patterns are introduced into the graph in the form of edges, which enhances the modeling effect of entity-level features. By enumerating bridge entities, the model generates a set of implicit path features for entity pair, effectively enriching the input features of the classifier.

four such issues for language annotation related to the establishment of annotation labels and the treatment of languages with more fine-grained, more coarse-grained, and cross-cutting categories. They proposed a lattice-like architecture for the annotation categories, which remains both intuitive for annotators and faithful to typological insights. In terms of temporal-textual embedding, Alonso et al. [41] suggested that there are three forms of temporal information representation: an explicit reference, an implicit reference, and a relative reference. Najafipour et al. [42] devised a neural network-based temporal-textual framework that generates tightly connected author subgraphs from microblog short-text contents. This model calculates the relevance score (edge weight) between authors by considering a combination of contents and concepts. Hosseini et al. [43] proposed a sophisticated framework that integrates a neural network-based time-aware word embedding component to collectively construct word vectors using infinite latent temporal facets. An elaborate generative model is employed to calculate edge weights based on heterogeneous temporal attributes.

Additionally, while most RE studies focus on triplets with a subject–predicate–object (SPO) structure, there are other works that incorporate a fourth dimension. Li et al. [44] introduced a quadruplet structure based on the triplet structure, recognizing the importance of SPO properties for ranking related entities and utilizing the knowledge graph. Chen et al. [45] proposed a deep quadruplet network with quadruplet loss for person re-identification, which results in a model output with greater inter-class variation and reduced intra-class variation compared to the triplet loss. Son et al. [46] designed a quadruplet convolutional neural network for multi-object tracking. This model learns to associate object detections across frames using quadruplet losses, which consider target appearances with their temporal adjacencies to improve data association.

3. Model

3.1. Problem definition

Assume that document $D = \{S_1, S_2, \dots, S_{N_s}\}$ consists of N_s unstructured word sequences and that set $E = \{e_1, e_2, \dots, e_{N_e}\}$ contains all entities in the document. The purpose of DocRE is to predict one or more predefined semantic relations between each set of head and tail entity pairs (e_h, e_t) in the document and output a relation subset $r_{h,t} \in \{R \cup \{None\}\}$, where $h, t \in \{1, 2, \dots, N_e\}$ and $h \neq t$, R represents the sum of the predefined relation types. The purpose of DocRE is to predict one or more predefined semantic relations between the head and tail entity (e_h, e_t) in each entity pair. In this study, the model expresses the extracted relations between the head and tail entity by producing a relation subset $r_{h,t} \in \{R \cup \{None\}\}$, where $h, t \in \{1, 2, \dots, N_e\}$ and $h \neq t$, R represents the set of predefined relation types. *None* is a classification label defined outside the set R and aimed to signify the absence of a predefined relation type between the head and tail entity. Incorporating such labels enables the model to convert RE from an N_r binary classification problem to a multi-classification challenge, where N_r denotes the aggregate number of predefined relation types. Each sentence in the document can be represented as $S_i = \{w_1^i, w_2^i, \dots, w_N^i\}$, where N represents the number of words in the sentence. Since an entity may have multiple mentions in the document, this paper defines the mention set of entity e_i as $M_i = \{m_1^i, m_2^i, \dots, m_{N_m}^i\}$, where N_m represents the total number of mentions of this entity in the document.

3.2. Hierarchical dependency tree construction and updates

For DocRE, there are four hierarchies of information in a document: the global features of document, the local features of sentences, the semantic features of entities and the contextual features represented by context. Previous studies tended to model

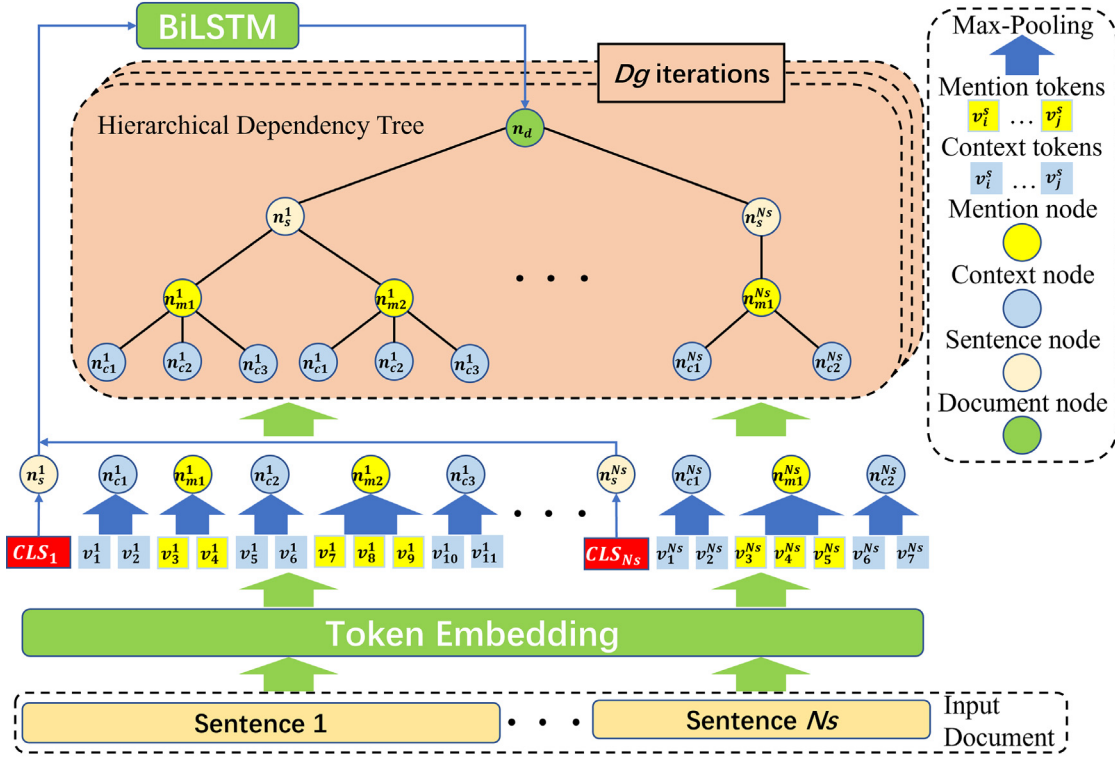


Fig. 3. Overall framework of the HDT model. The model first generates node features at each hierarchy, including context nodes, mention nodes, sentence nodes and document node. Then, an adjacency matrix is constructed based on the co-occurrence and co-referential relations between nodes to add edge information to the graph network. Finally, multiple rounds of aggregation operations are performed based on the GCN to complete fine-grained high-dimensional feature extraction.

the entire document at once and focus on extracting the dependency structure among entities. This coarse-grained modeling method not only loses the local features and hierarchical concepts of a sentence but also fails to effectively use contextual information to improve the precision of relation prediction. This paper proposes a fine-grained reconstruction method for document based on hierarchical dependencies. The method captures the structural dependence features between each hierarchy of information in units of sentences to generate a Hierarchical Dependency Tree (HDT) of the document, and it performs the aggregation operation of the tree diagram based on the graph convolutional network (GCN) to complete the feature update.

The first step of the model executes the feature embedding of the input document, effectively imitating many successful natural language processing (NLP) cases. This work is based on static word embedding technology and BERT to implement two sets of feature encoders. The specific details are presented in the subsequent section on the experimental settings. Different from the previous work, our model abandons the idea of embedding the entire document at one time and instead builds the representation mapping sentence-by-sentence. Conceptualizing of the sentence as a unit effectively reduces the sequence length of a single embedding, thereby avoiding the problem of passive information clipping. The embedded expression of each sentence in the document is as follows:

$$S_i' = \text{Token_Embedding}(S_i) = \{v_1^i, v_2^i, \dots, v_N^i\} \quad (1)$$

where $i \in \{1, 2, \dots, N_s\}$, BiLSTM or BERT is chosen as $\text{Token_Embedding}(\cdot)$ in this work.

After obtaining the feature sequence of a document, the model divides the corresponding context region by the border of the mentions. Max-pooling is applied to the feature sequence corresponding to the mention and the context region to initialize the corresponding node features.

Using the first sentence in Fig. 3 as an example, this sentence contains two mentions, $\text{mention}_1 = \{v_3^1, v_4^1\}$ and $\text{mention}_2 = \{v_7^1, v_8^1, v_9^1\}$, and it can be divided into three context information regions, $\text{context}_1 = \{v_1^1, v_2^1\}$, $\text{context}_2 = \{v_5^1, v_6^1\}$ and $\text{context}_3 = \{v_{10}^1, v_{11}^1\}$. The model implements the max-pooling of these five subsequences to generate the corresponding mention node and context node as follows:

$$\begin{aligned} n_{mi}^1 &= \text{MaxPool}(\text{mention}_i) \\ &= \text{MaxPool}(\{v_j\}_{j=s}^e) \end{aligned} \quad (2)$$

$$\begin{aligned} n_{ci}^1 &= \text{MaxPool}(\text{context}_i) \\ &= \text{MaxPool}(\{v_j\}_{j=s}^e) \\ &= \{\max(\{v_{jk}\}_{j=s}^e)\}_{k=1}^d \end{aligned} \quad (3)$$

where n_{mi}^s represents the node feature of the i th mention in the s th sentence of the document and n_{ci}^s denotes the node feature of the i th context region in the s th sentence. v_k is a word representation in the subsequence, s and e represent the starting and ending positions of the subsequence in the s th sentence. d is the feature dimension after token embedding. Fig. 4 shows the max-pooling calculation schematic diagram. In this work, the dimension of the word embedding is the same, and thus, the max-pooling layer can convert the feature sequence of any length into a feature vector of uniform dimension.

After all intra-sentence node modeling is finished, the model uses the feature of the sentence-initial identifier CLS_j as the node feature n_s^j of each sentence S_j in the HDT, and it builds BiLSTM to generate the global node feature of the document by integrating the feature sequence of the sentence nodes as follows:

$$\begin{aligned} C_j &= \sigma(W_f[n_s^j; h_{j-1}] + b_f) \times C_{j-1} + \\ &\quad \sigma(W_i[n_s^j; h_{j-1}] + b_i) \times \tanh(W_c[n_s^j; h_{j-1}] + b_c) \end{aligned} \quad (4)$$

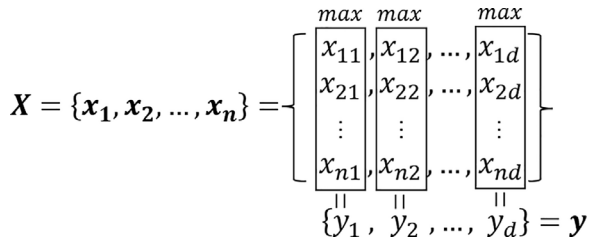


Fig. 4. Max-pooling calculation schematic diagram. n indicates the length of the input feature sequence, d represents the embedded dimension of each word in the feature sequence, and $\mathbf{y} \in \mathbb{R}^d$ indicates the feature vector of the output by the max-pooling layer.

$$h_j = \sigma(W_o[n_s^j; h_{j-1}] + b_o) \times \tanh(C_j) \quad (5)$$

where $j \in \{1, 2, \dots, N_s\}$ and $\sigma(\cdot)$ is the sigmoid function. $[x; y]$ represents the vector that concatenates on the feature dimension, $W_{f/i/c/o}$ and $b_{f/i/c/o}$ represent all learnable parameters in BiLSTM, h_j is the hidden state generated by the BiLSTM modeling the j th input sequence, and C_j denotes the memory cell maintained by the network in this time. We use the hidden state of the last moment (i.e., $j = N_s$) as the feature representation of the document node, denoted by n_d .

When all node modeling is finished, the adjacent matrix of the HDT is built with the following rules: the document node n_d is adjacent to all sentence nodes, the sentence node n_s^j is adjacent to the intra-sentence mention nodes, and the mention node n_{mi}^j is adjacent to the context nodes located in the same sentence. The HDT realizes the information interaction between nodes by using the adjacent matrix, ensuring that each hierarchy information participates in the deep reasoning process of the graph network with different weights.

At this point, the document of the chain structure is restructured into an HDT, and the nodes of each hierarchy in the HDT can realize the interaction and aggregation of neighborhood node information, completing the feature update. The model designs the GCN layers to perform the above aggregation and update operation and apply residual connection [47] and layer normalization [48] between the layers. It is worth noting that there are three types of edges in the HDT. The model sets independent parameters for different aggregate information sources during the convolution calculation, and the corresponding mathematical expression is expressed as follows:

$$\begin{aligned} \bar{n}_i^l = & \text{ReLU}\left(\sum_{j \in N_{cm}(i)} (W_{cm} n_j^l) + \sum_{j \in N_{ms}(i)} (W_{ms} n_j^l) \right. \\ & \left. + \sum_{j \in N_{sd}(i)} (W_{sd} n_j^l) + b_h\right) \end{aligned} \quad (6)$$

$$\begin{aligned} n_i^{l+1} = & \text{LayerNorm}(n_i^l + \bar{n}_i^l) \\ = & \frac{(n_i^l + \bar{n}_i^l) - E(n_i^l + \bar{n}_i^l)}{\sqrt{\text{Var}(n_i^l + \bar{n}_i^l) + \epsilon}} \end{aligned} \quad (7)$$

to facilitate understanding, the nodes in the HDT are summarized to $\{n_1, n_2, \dots, n_{N_h}\}$, where N_h represents the total number of nodes in the HDT. n_i^l represents the feature representation of the i th node in the HDT after l th layers of the GCN, and $\text{ReLU}(\cdot)$ is a non-linear activation function. $N_{cm}(i)$, $N_{ms}(i)$ and $N_{sd}(i)$ represent the subsets from different hierarchies of nodes in neighboring of node n_i . $W_{cm/ms/sd} \in \mathbb{R}^{d \times d}$ and $b_h \in \mathbb{R}^d$ are learnable parameters in the model, and d is the dimension of the word embedding. The layer normalization formula comprises $E(\cdot)$ for mean calculation, $\text{Var}(\cdot)$ for variance calculation, and ϵ as a small value added to prevent division by zero.

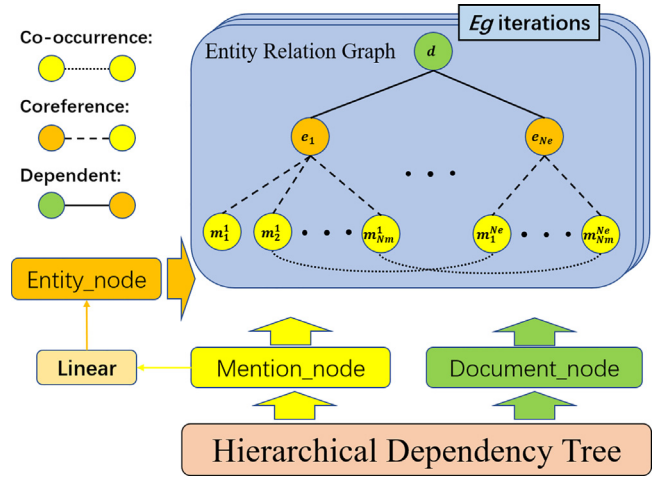


Fig. 5. Execution flow of the entity correlation graph (ECG). First, the model initializes the entity nodes based on the features of the mention nodes output by the HDT. Then, edges are added between entity nodes, mention nodes and document node by co-occurrence relation, co-reference relation and structural dependence. Finally, the model GCN layers perform aggregation, update the ECG, and output the final feature representation of the entity nodes.

After several rounds of aggregation and updates, the HDT integrates the modeling results of the mention nodes and document node of all GCN layers as the important features of subsequent deep reasoning. Assuming that the number of GCN layers is set to d_g , the model concatenates the features calculated by the convolution of each layer, and a linear layer is designed to maintain the same dimension between the output and the input.

$$n_i^l = W_{go}[n_i^0; n_i^1; \dots; n_i^{d_g}] + b_{go} \quad (8)$$

where n_i^l is the final feature representation of node n_i in the HDT, $[x; y]$ represents the vector concatenation on the feature dimension, and $W_{go} \in \mathbb{R}^{(d_g+1)d \times d}$ and $b_{go} \in \mathbb{R}^d$ are learnable model parameters.

3.3. Entity feature modeling

The performance of the RE model is highly related to the quality of the entity feature modeling. This section elaborates on the modeling process of entity features based on the HDT node information.

First, the model uses the mention nodes output by the HDT to initialize the feature of the entity node. All mention nodes corresponding to the entity in the document are accumulated, and the initial features of the entity node are generated through a fully connected layer:

$$e_i^0 = W_m \sum_{j=1}^{N_m} m_j^i + b_m \quad (9)$$

where i represents the entity number and N_m represents the total number of mentions for entity e_i in the document. m_j^i represents the node feature of the j th mention for entity e_i . $W_m \in \mathbb{R}^{d \times d}$ and $b_m \in \mathbb{R}^d$ are learnable parameters, optimizing with the training of the model.

Then, on the basis of the initial entity node, the model, combined with the mention nodes and document node output by the HDT, jointly builds an entity correlation graph (ECG). The composition rules are as follows: (i) Set edges according to the co-occurrence relation (appearing in the same sentence) between mention nodes. (ii) According to the coreference relation (referring to the same entity) between the entity node and mention

nodes to set edge. (iii) Set the edge between document node and all entity nodes.

Finally, the ECG has three types of nodes and three types of edges, as shown in Fig. 5. The figure makes an explicit distinction between different types of nodes and edges. The GCN layer executes the aggregation and update of the ECG and also applies residual connection and layer normalization between each layer. The corresponding detailed mathematical expression is consistent with the HDT modeling process. For details, please refer to Formula (6) to Formula (7).

Assuming that the number of GCN layers on the ECG is E_g , the model concatenates the output of the entity nodes and document node on each layer and generates the final feature representation through a fully connected layer:

$$e'_i = W_e[e_i^0; e_i^1; \dots; e_i^{E_g}] + b_e \quad (10)$$

$$d'_i = W_d[d_i^0; d_i^1; \dots; d_i^{E_g}] + b_d \quad (11)$$

where $[x; y]$ indicates vector concatenates on the feature dimension and $W_e, W_d \in \mathbb{R}^{(E_g+1)d \times d}$, $b_e, b_d \in \mathbb{R}^d$ are learnable parameters. e'_i and d'_i represent the feature representation of the entity e_i and the document with deep modeling.

Since the mention nodes and document node in the HDT have aggregated high-dimensional features at different hierarchies, the updated entity nodes in the ECG include the contextual features of all hierarchies in the document.

3.4. Bridge path enhanced relation extraction

For the final RE stage, we propose a modeling method for the bridge path of the entity pair. The classifier of the model introduces bridge path features on the basis of the node features of the entity and document to improve the predictive performance.

Suppose the entity pair is represented by (e_h, e_t) . The model proposes the definition of relational bridge entities as follows: if the mention set of entity e_q has a mention that both satisfies the co-occurrence relation with the mentions of head entity e_h and tail entity e_t , the entity e_q is a relational bridge entity of the entity pair of (e_h, e_t) .

First, the model enumerates all relational bridge entities for entity pair (e_h, e_t) according to the above definition and builds a set denoted by E_q . Each relational bridge entity can form a bridge path with the entity pair, such as e'_h, e'_q, e'_t . To avoid $E_q = \emptyset$, the document node is the default relational bridge entity of all entity pairs. Therefore, the node set of the relational bridge entity can be represented by $E'_q = \{E_q \cup \{d'\}\}$. Since the bridge path is essentially a hidden information channel for the entity pair in the document, the model uses BiLSTM, which is good at serial modeling, to calculate the feature of each bridge path as follows:

$$C_j = \sigma(W_{bf}[e'_j; h_{j-1}] + b_{bf}) \times C_{j-1} + \sigma(W_{bi}[e'_j; h_{j-1}] + b_{bi}) \times \tanh(W_{bc}[e'_j; h_{j-1}] + b_{bc}) \quad (12)$$

$$h_j = \sigma(W_{bo}[e'_j; h_{j-1}] + b_{bo}) \times \tanh(C_j) \quad (13)$$

where $j \in \{h, q, t\}$ and $\sigma(\cdot)$ is the sigmoid function. $[x; y]$ represents vector concatenation on the feature dimension, and $W_{bf/bi/bc/b_o}$ and $b_{bf/bi/bc/b_o}$ represent all learnable parameters in the BiLSTM. h_j is the hidden state generated by the BiLSTM modeling of the j th input sequence, and C_j denotes the memory cell maintained by the network at this time. We use the hidden state of the last moment (i.e., $j = t$) as the feature representation of each bridge path, denoted as $b_{h,t}^q$.

Then, the features of multiple bridge paths between an entity pair are merged by weighted summation. Inspired by the alignment scheme of the self-attention mechanism, the model design includes the following formula, which completes the weight

calculation of each path and information integration of all path features as follows:

$$b_{h,t} = \text{softmax}\left(\frac{(e'_t + e'_h)B_{h,t}^T}{\sqrt{d}}\right)B_{h,t} \quad (14)$$

where $B_{h,t} \in \mathbb{R}^{N_q \times d}$ represents the vector sequence of all bridge path features, N_q is the number of relational bridge entities, and $b_{h,t}$ is the final representation of integrating the features of all bridge paths for the entity pair (e_h, e_t) .

Finally, the classifier of the model uses the features of the entity pair, document and bridge path as the input execution DocRE and outputs the predictive score of each type of relation between the entity as follows:

$$r_{h,t} = \sigma(W_2[(W_1[e'_h; b_{h,t}; e'_t] + b_1); d'] + b_2) \quad (15)$$

where $\sigma(\cdot)$ is the sigmoid function, and $[x; y]$ represents vector concatenation on the feature dimension. $W_1 \in \mathbb{R}^{3d \times d}$, $W_2 \in \mathbb{R}^{2d \times n_{re}}$, $b_1 \in \mathbb{R}^d$ and $b_2 \in \mathbb{R}^{n_{re}}$ are learnable parameters of the model. n_{re} represents the total number of predefined relation types. Each dimension of the output vector $r_{h,t} \in \mathbb{R}^{n_{re}}$ corresponds to a predicted score for the existence of such a relation between the entity pair (e_h, e_t) .

3.5. Loss function

Since there may be multiple types of relations between an entity pair, this work treats DocRE as a multiple binary classification problem. Binary cross-entropy (BCE) is set as the loss function of the model to calculate the prediction error.

Suppose $r_{h,t}^i \in [0, 1]$ is the real relation label in the document, and it represents whether there is a real relation of type i between entity pair (e_h, e_t) . N_e represents the total number of entities, and n_{re} is the number of predefined relation types. The loss function can be defined as follows:

$$\text{loss} = \sum_{h=1}^{N_e} \sum_{t=1, t \neq h}^{N_e} \sum_{i=1}^{n_{re}} (\text{BCE}(r_{h,t}^i, \overline{r_{h,t}^i})) \quad (16)$$

where $r_{h,t}^i$ represents the value of the model result $r_{h,t}$ in the i th dimension, that is, the prediction score of the i th type of relation between the entity pair. Based on the prediction error loss of the loss function output, the model calculates the gradient value of each parameter through back propagation and the chain rule, which supports the optimizer to complete the update of model parameters.

4. Experiments

4.1. Datasets

This work used the DocRED [10] dataset to evaluate the model performance. This dataset is the largest human-annotated dataset currently used for DocRE, and the documents were taken from Wikipedia and Wikidata. The DocRED dataset contains 132,375 entities, 63,427 relation instances, 56,354 relational facts and 96 relation types. Each document contains an average of 19.5 entities, and the average number of mentions of each entity is 1.4. More than 40% of the relations in the document need to read multiple sentences for cross-sentence reasoning, and the existence of a large number of inter-sentence relational facts is a significant difference from the sentence-level RE dataset [49–51]. To fairly compare the performance with previous works, the experiment strictly followed the standard partition format of the DocRED dataset in which 3053 of the documents were used for model training, 1000 documents were used as the development set, and 1000 documents were used as the test set. The statistics of the DocRED dataset are shown in Table 1.

Table 1

Statistics for the DocRED dataset. (Doc: document, Sen: sentence, En: entity, Re-t: relation type, Re-ins: relation instance, Re-f: relational fact).

Dataset	Doc	Sen	En	Re-t	Re-ins	Re-f
Total	5053	40276	132375	96	63427	56354
Training Set	3053	–	–	96	38269	34715
Dev Set	1000	–	–	96	12332	11790
Test Set	1000	–	–	96	12842	12101

Table 2

Hyperparameter settings.

Hyperparameter	Notation	Value
Word embedding dimension	d	512/768
GCN layers in HDT	D_g	2
BiLSTM layers in HDT	bl_{doc}	2
GCN layers in ECG	E_g	2
BiLSTM layers in path modeling	bl_{path}	2
Optimizer	–	Adam
Learning rate	lr	$5e-4$
Dropout value	–	0.3
Batch size	–	16/4
Max epoch	–	100
Early-stopping epoch	–	20

4.2. Experimental setup

In this work, two groups of text feature embedding schemes were implemented: (i) The model directly used the standard static word vector file provided by the DocRED dataset, and the BiLSTM further modeled the embedded feature sequence to expand the feature dimension. (ii) The model used the BERT_{base} provided by Wolf et al. [52] to generate context-dependent dynamic word vectors for input documents. The text feature dimension d of the static word embedding and dynamic word embedding were 512 and 768, respectively. The number of GCN layers D_g and E_g on the HDT and ECG were both set to 2, and the number of BiLSTM layers bl_{doc} and bl_{path} used to model the features of the document nodes and bridge paths were also set to 2. The model used Adam [53] as the parameter optimizer, and the learning rate lr was set to $5e-4$. A dropout layer with a value of 0.3 was applied to the output of each sublayer in the model to prevent overfitting during training; during the training process, the batch size of the BERT-based model was set to 4, and that of the model based on the static word vector was set to 16. The maximum training epoch was 100, and the early-stopping epoch was set to 20. The model was trained on the training set based on the above hyperparameter settings, and the DocRE performance was verified by combining offline (development set) and online (test set) methods. The above hyperparameters are summarized in Table 2. The experiment was based on PyTorch-1.2 to build the deep learning algorithm framework. The CPU model of the experimental platform was an Intel i7-9700K, and the GPU model was an NVIDIA TITAN-RTX.

4.3. Performance metrics

The widely used metrics, precision, recall and F1 score, were used in our experiments. Precision (also called the positive predictive value) represents the ability of a model to identify only the relevant samples. Recall (also known as sensitivity) represents the ability of a model to find all of the relevant samples within a dataset. Since precision and recall are two contradictory indexes, one single metric is insufficient for imbalanced datasets. The experiment also used the standard F1 score, which can simultaneously weigh the precision and recall to evaluate the DocRE performance of the model.

In a classification task, the precision for a class is the number of true positives divided by the total number of elements labeled as belonging to the positive class (i.e., the sum of true positives and false positives). Recall, in this context, is defined as the number of true positives divided by the total number of elements that actually belong to the positive class (i.e., the sum of true positives and false negatives). Mathematically, a true positive (TP) represents the number of items correctly labeled as belonging to the positive class, a false positive (FP) represents the number of items incorrectly labeled as belonging to the positive class, a true negative (TN) represents the number of items correctly labeled as belonging to the negative class, and a false negative (FN) represents the number of items that were not labeled as belonging to the positive class but should have been. Finally, the F1 score is the organic unity of the precision and recall, and it is expressed mathematically as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

In addition, some relational facts present in both the training and development/test sets may cause evaluation bias. The model may memorize the pattern features of these relational facts during training and improve the prediction performance on the development/test sets by using prior knowledge. However, overlap in relational facts between the training and development/test sets is inevitable since many common relational facts are likely to be shared in different documents. In view of the above phenomena, based on the global F1 score reported in the experiment, we also report the F1 scores excluding those relational facts shared by the training and dev/test sets, denoted as Ign-F1.

4.4. Baseline systems

We chose some baseline models from previously published papers on the DocRED dataset. In general, these models can be divided into two categories: GloVe-based models and BERT-based models, both of which provide embeddings as representations of words. The former uses non-contextual and fixed embeddings, while the latter uses contextual and dynamic embeddings. Based on the aforementioned categorization, we compare our model with the following published models:

Yao et al. [10] proposed models to encode documents as a sequence of hidden state vectors using the **CNN**, **LSTM**, and **BiLSTM** encoders to predict entity relations based on their representations. They also introduced the **Context-Aware** model, which is a method that they created for DocRED based on work by Sorokin and Gurevych [54], this previous work uses context-aware BiLSTM to encode text and incorporate attention mechanisms to capture contextually-dependent relational information during prediction.

HIN and **HIN_{BERT}** [55] introduced the use of a hierarchical inference network for aggregating information from entities to sentences and on to the document level, enabling semantic decision-making across the document. GloVe or BERT is used to generate word embeddings in this work.

RE_{BERT} and **Two-Phase_{BERT}** [56] are based on the fine-tuned BERT. While the former is a straightforward pipeline model, the latter uses a two-step strategy. During the first phase, Two-Phase_{BERT} generates Boolean predictions about the relationship between entity pairs, and during the second phase, it predicts the final relation. Other pre-trained models, such as **Coref_{BERT}** [22], have also been used as encoders for the DocRE task.

Nan et al. [18] proposed **LSR** and **LSR_{BERT}**, both of which create an induced latent graph structure illustrating information propagation. These models facilitate the incremental aggregation of relevant information for logical reasoning. Nan et al.

also adapted three advanced graph-based models, including the **GAT** [57], **GCNN** [19] and **EoG** [20] for RE on DocRED, either using homogeneous or heterogeneous graphs based on the entire document.

Furthermore, more recent advanced graph networks are listed as comparison methods for the proposed model in this paper. The **AGGCN** [34], an intra-sentence graph-based RE model that uses dependency trees, is enhanced by introducing attention guide graph convolutional networks, which select relevant substructures for RE. Li et al. [58] proposed **GEDA** and **GEDA_{BERT}**, which use a graph-based dual attention network to model complex interactions between sentences and possible relation instances. Wang et al. [2] proposed **DISCO**, which aims to select relevant evidence and demonstrate the reasoning process. This model uses a document-graph to indicate valid semantic associations between multiple text units and incorporates reasoning modules based on rhetorical structure theory to capture relevant evidence. **SSAN_{BERT}** [59] leverages co-occurrence information between entity mentions, using distantly supervised data for pretraining. **ESA** [27] uses inter-sentence attention to select essential features at both the sentence and document levels, which are then combined with document gating for a more complete understanding. **E2GRE** [29] generates entity-guided sequences for input into a language model and utilizes internal attention from the last layers to focus on relevant areas of a document.

5. Results and analysis

In this section, we first report the overall test results of the proposed model and several advanced models on the DocRED dataset and provide a comparative analysis. Then, we discuss the conducted performance tests for inter-sentence relation and intra-sentence relation, analyzing the differences among them. To explore the influence of the number of sentences in the document on the model performance, the documents in the development set were divided into multiple subsets according to the different number of sentences to test the model performance. The hyperparameter experiment explores the influence of different hyperparameter settings on the model performance, and the HDT pruning experiment reflects the importance of each hierarchical feature for model reasoning. Finally, an ablation experiment was performed to verify the effectiveness of each component of the proposed model.

5.1. Quantitative results

Tables 3 and 4 present the performance comparison between the HDT-BP and other advanced models on the DocRED dataset. The models in Table 3 all use static word embedding techniques to generate feature representations of words. The models in Table 4 all implement dynamic word embeddings based on BERT_{base}. To distinguish between models, the latter is displayed with subscripts, such as HDT-BP_{BERT}.

Without using the PLM, the HDT-BP consistently leads the other comparison models on the development set and test set with 55.36% and 54.80% F1 scores. By observing Table 3, it can be seen that among the sequence-based models, the model with the ability to capture contextual information exhibits better performance. For example, the performance of LSTM is better than CNN. BiLSTM also outperforms LSTM due to its ability to capture bidirectional context dependencies. Compared with the sequence-based models in Table 3, the HDT-BP achieves an F1 performance improvement of 2.41% ~ 11.91% on the development set and also assumes the leading position on the test set with a performance advantage of 1.50% ~ 12.54%. The results demonstrate that the introduction of the graph structure is beneficial to the model

Table 3

Results of different models on the DocRED dataset (w/o BERT_{base}).

Model	Development set		Test set	
	F1	Ign-F1	F1	Ign-F1
Sequence-based Models				
CNN [10]	43.45	41.58	42.26	40.33
LSTM [10]	50.68	48.44	50.07	47.71
BiLSTM [10]	50.94	48.87	51.06	48.78
Context-Aware [10]	51.09	48.94	50.70	48.40
HIN [55]	52.95	51.06	53.30	51.15
Graph-based Models				
GAT [57]	51.44	45.17	49.51	47.36
GCNN [19]	51.52	46.22	51.62	49.59
EoG [20]	52.15	45.94	51.82	49.48
AGGCN [34]	52.47	46.29	51.45	48.89
GEDA [58]	53.60	51.03	52.97	51.22
LSR [18]	55.17	48.82	54.18	52.15
HDT-BP(Ours)	55.36	53.27	54.80	52.40

Table 4

Results of different models on the DocRED dataset (w/ BERT_{base}).

Model	Development set		Test set	
	F1	Ign-F1	F1	Ign-F1
w/ BERT _{base}				
RE _{BERT} [56]	54.16	–	53.20	–
Two-Phase _{BERT} [56]	54.42	–	53.92	–
HIN _{BERT} [55]	56.31	54.29	55.60	53.70
GEDA _{BERT} [58]	56.16	54.52	55.74	53.71
Coref _{BERT} [22]	57.51	55.32	56.96	54.54
DISCO [2]	57.78	55.91	55.70	55.01
E2GRE [29]	58.72	55.22	–	–
ESA [27]	58.28	56.20	58.04	55.71
SSAN _{BERT} [59]	59.19	57.03	58.16	55.84
LSR _{BERT} [18]	59.00	52.43	59.05	56.97
HDT-BP _{BERT} (Ours)	59.37	57.17	58.77	56.28

in terms of understanding the document more deeply. Although sequence-based models can capture certain lengths of context dependencies, they are not sensitive to the feature information of local structure. In the performance comparison between graph-based models, the F1 score of the HDT-BP is slightly better than the advanced method LSR [18] (+0.19%, +0.62%) on the development set and test set, but it still has obvious advantages in terms of the Ign-F1 score (+4.45%, +0.35%). Compared with other graph-based models, the HDT-BP demonstrates significant performance improvement, with the maximum performance improvement on the development set and test set being 3.92% and 5.29%, respectively. An explanation of this advantage stems from the fine-grained modeling of document by the HDT-BP. Different from other graph-based models that use the whole document as the modeling unit, the HDT-BP captures the high-dimensional features of each hierarchy in the sentence units, and the features of each hierarchy play an irreplaceable role in deep reasoning.

Observation of Table 4 reveals that the performance of the same model appears to be significantly improved after joining BERT_{base}. Compared with the HDT-BP, HDT-BP_{BERT} achieves 4.01% and 3.97% improvements in terms of the F1 score on the development and test sets. This phenomenon indicates that dynamic word embedding is beneficial to better modeling and reasoning by graph networks. It is worth noting that the performance of the HDT-BP is still better than RE_{BERT} [56] (+1.20%, +1.60%) and Two-Phase_{BERT} [56] (+0.94%, +0.88%) without using BERT_{base}. Compared with the previous advanced model LSR_{BERT} [18], the F1 score of HDT-BP_{BERT} on the development set and test set is slightly improved (+0.37%, +0.28%), but the Ign-F1 score on the development set is significantly improved (+4.74%).

Table 5
Performance of inter/intra-sentence RE.

Model	w/o BERT _{base}		w/ BERT _{base}	
	Intra-F1	Inter-F1	Intra-F1	Inter-F1
CNN [18]	51.87	37.58	–	–
LSTM [18]	56.57	41.47	–	–
BiLSTM [18]	57.05	43.49	–	–
Context-Aware [18]	56.74	42.26	–	–
GCNN [19]	57.78	44.11	–	–
EoG [20]	58.90	44.60	–	–
GAT [57]	58.14	43.94	–	–
AGGCN [34]	58.76	45.45	–	–
GEDA [58]	58.83	47.72	–	–
LSR [18]	60.83	48.35	–	–
HDT-BP(Ours)	61.59	48.18	–	–
RE _{BERT} [56]	–	–	61.61	47.15
Two-Phase _{BERT} [56]	–	–	61.80	47.28
GEDA _{BERT} [58]	–	–	61.85	49.46
LSR _{BERT} [18]	–	–	65.26	52.05
HDT-BP _{BERT} (Ours)	–	–	66.29	51.36

5.2. Analysis of inter/intra-sentence RE

The biggest difference between DocRE and sentence-level RE lies in the existence of inter-sentence relations in a document. To explore the performance difference of the model on inter-sentence and intra-sentence RE, the F1 scores related to these tasks are calculated in the experiment and reported in Table 5. Experimental definition: If there is no co-occurrence relation between the two mentioned sets corresponding to the head and tail entity of a relation, the relation is defined as an inter-sentence relation. In Table 5, Inter-F1 represents the F1 score of inter-sentence RE, and Intra-F1 reflects the F1 score of intra-sentence RE. It can be observed that the performance of inter-sentence RE for all models is significantly lower than that of intra-sentence RE, which proves that the difficulty of inter-sentence RE is far greater than that of intra-sentence RE. The proposed model achieves a competitive F1 score (48.18%, 51.36%) in terms of inter-sentence RE, and the maximum performance improvement is 10.6% and 4.21%. In terms of intra-sentence RE, the F1 score of the HDT-BP is significantly better than other comparison models, whether based on static word embeddings (+0.76% ~ +9.72%) or BERT_{base} (+1.03% ~ +4.68%). It is worth mentioning that the performance in terms of inter-sentence RE by the HDT-BP is better than RE_{BERT} [56] and Two-Phase_{BERT} [56] without using BERT_{base}.

5.3. Analysis of the number of sentences

To further explore the influence of the number of sentences contained in a document on the model performance, document instances from the DocRED development set were divided into four sub-datasets according to the number of sentences: $\{0 < n_{sen_num} \leq 5\}$, $\{5 < n_{sen_num} \leq 10\}$, $\{10 < n_{sen_num} \leq 15\}$, $\{n_{sen_num} > 15\}$. Fig. 6 reports the F1 score and Ign-F1 score of the HDT-BP and baseline model on each sub-dataset, where LSR [18] was set as the baseline model. Observing the results in the figure, we can draw the following conclusions: First, with the increase in the number of sentences contained in the document, the F1 score and Ign-F1 score of the HDT-BP and LSR both exhibit a significant downward trend. This phenomenon indicates that the difficulty of RE is positively correlated with the number of sentences in the document and also proves that DocRE is more challenging than sentence-level RE. Second, the performance loss indicated by the F1 score and Ign-F1 score between the head and tail sub-datasets of the HDT-BP is 5.54% and 5.89%, and that of LSR is 7.23% and 7.45%. These results demonstrate that the

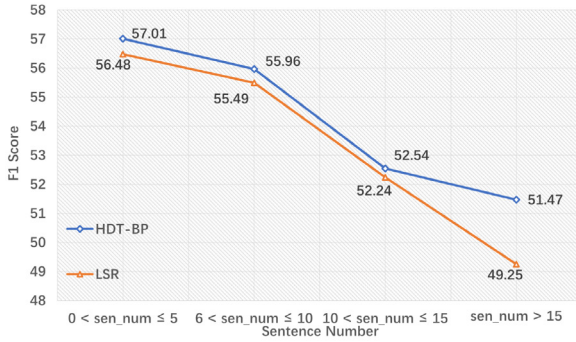
Table 6
Performance with different hyperparameter settings.

HDT-BP _{BERT}			F1	Ign-F1
D_g	bl_{doc}	bl_{path}		
1	1	1	58.95	56.63
2	1	1	58.70	56.49
3	1	1	58.63	56.50
1	2	1	58.61	56.30
2	2	1	58.92	56.72
3	2	1	58.69	56.60
1	1	2	58.68	56.44
2	1	2	58.38	56.26
3	1	2	58.86	56.63
1	2	2	58.24	56.07
2	2	2	59.37	57.17
3	2	2	59.05	56.84
1	1	3	58.58	56.39
2	1	3	58.94	56.68
3	1	3	58.67	56.49
1	2	3	58.67	56.48
2	2	3	58.68	56.49
3	2	3	58.87	56.62

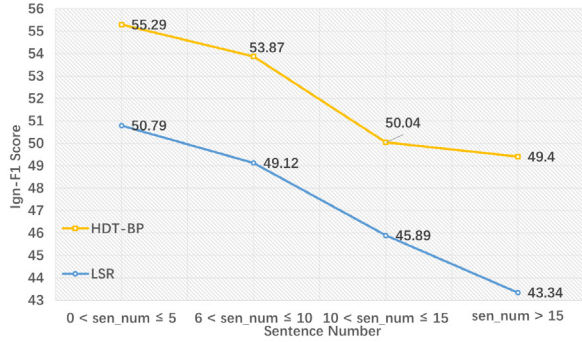
performance loss of the proposed model with an increase in the number of sentences is significantly smaller than that of the baseline model, exhibiting stronger generalization performance for different sub-datasets. Third, the results in Fig. 6(a) show that the F1 scores of the HDT-BP on each sub-dataset are better than those of LSR. When the number of sentences is less than 15, the performance advantage of the HDT-BP is within 1%. When the number of sentences is greater than 15, the F1 score of the HDT-BP is 51.47%, which is 2.22% higher than that of the baseline model. Finally, Fig. 6(b) depicts the Ign-F1 score of the two models on each sub-dataset, where the HDT-BP outperforms the baseline model by a significant performance advantage on each sub-dataset. The smallest performance advantage occurs on the sub-dataset $\{10 < n_{sen_num} \leq 15\}$, where the Ign-F1 score of the HDT-BP is 4.15% higher than that of the baseline model. When the number of sentences in the document is greater than 15, the HDT-BP demonstrates the largest performance advantage (+6.06%) and achieves the leading result on this sub-dataset with a 49.40% Ign-F1 score. This experiment verifies the effectiveness of the model when handling documents with different numbers of sentences, and it proves that the HDT-BP has obvious performance advantages when extracting relational facts that did not appear in the training set.

5.4. Analysis of hyperparameter

This experiment mainly explores the influence of three hyperparameters on the model performance: the number of GCN (D_g) and BiLSTM (bl_{doc}) layers in the HDT and the number of BiLSTM (bl_{path}) layers in the bridge path modeling. Table 6 shows the test results of HDT-BP_{BERT} on the DocRED development set with 18 different hyperparameter settings. It can be observed from the table that the model achieves the optimal performance when all three hyperparameters are set to 2. When both bl_{doc} and bl_{path} are set to 2, too few or too many GCN layers may lead to inadequate modeling or overfitting during the training process, resulting in performance degradation. Similarly, when D_g and bl_{doc} are both 2, the model performance is not positively correlated with the size of the parameter bl_{path} . In addition, increasing or decreasing the number of BiLSTM layers leads to a model performance decrease. The above analysis proves that the hyperparameter settings of the proposed model are reasonable and effective.



(a) F1 score for different sentence numbers.



(b) Ign-F1 score for different sentence numbers.

Fig. 6. F1/Ign-F1 score for different number of sentences.

5.5. HDT pruning study

The model proposed in this paper reconstructs the chain structure into an HDT according to the hierarchical dependency relation in the document, and the HDT is aggregated and updated to complete DocRE. To explore the contribution of the features of each hierarchy in the HDT during the model inference process, we conducted an experiment to prune the HDT and test the change in the model performance with the DocRED development set. The experimental results are shown in Fig. 7. First, the document node n_d representing the global feature at the top of the HDT was pruned (w/o Node_doc), and the F1 score and Ign-F1 score of the model were both reduced by 1.83%. Then, the sentence nodes n_s were pruned (w/o Node_sen), where the document node was directly connected to the mention nodes. The results show that pruning sentence nodes only results in a slight performance degradation (−0.56%, −0.63%). Finally, all context nodes n_c were pruned (w/o Node_con), and the F1 score and Ign-F1 score of the model on the DocRED development set were 54.34% and 52.20%. The above results indicate that document node makes the greatest contribution to the model, followed by context nodes, and sentence nodes have the least impact on the model performance. We analyze that this phenomenon is due to the large proportion of document node during the modeling process since they not only represent global features but also participate in bridge path modeling. The mention and context information within the sentence are finely segmented and modeled separately, so the sentence nodes have some redundant information. In addition, we also replace the BiLSTM in the HDT-generated document node with max-pooling. The F1 score and Ign-F1 score were reduced by 1.07% and 1.09%, respectively, which is greater than the influence of pruning sentence nodes.

5.6. Ablation study

We conducted four independent experiments on the DocRED development set to evaluate the contributions of different components in the model. Experimental results are shown in Table 7, where the ablation test of each component includes two sets of results: one uses BERT (w/ BERT) as token embedding and another uses BiLSTM (w/o BERT). This study involved several ablation experiments, beginning with the removal of the document feature in the classifier layer of the model (−doc_feature). The F1 and Ign-F1 scores of the HDT-BP are 54.48% and 52.28%, while these scores of HDT-BP_{BERT} are 58.76% and 56.52%. Next, we tested the model without using the bridge path feature (−bri_path), which resulted in a decrease in F1 score of 1.21%/0.87% and a decrease in Ign-F1 score of 1.42%/1.00% for both the HDT-BP and HDT-BP_{BERT}, respectively. In the third experiment, the model

Table 7

Ablation study on the DocRED development set.

Model		F1	Ign-F1	Inter-F1	Intra-F1
HDT-BP	w/o BERT _{base}	55.36	53.27	48.18	61.59
	w/ BERT _{base}	59.37	57.17	51.36	66.29
−doc_feature	w/o BERT _{base}	54.48	52.28	46.18	61.52
	w/ BERT _{base}	58.76	56.52	50.53	65.60
−bri_path	w/o BERT _{base}	54.15	51.85	45.91	61.45
	w/ BERT _{base}	58.50	56.17	49.98	66.10
−HDT	w/o BERT _{base}	53.20	51.05	44.99	60.02
	w/ BERT _{base}	57.76	55.59	48.83	65.94
−Both	w/o BERT _{base}	52.78	50.54	44.35	60.32
	w/ BERT _{base}	57.12	55.00	48.20	64.82
Replace GRU	w/o BERT _{base}	55.35	53.25	47.94	61.81
	w/ BERT _{base}	59.27	57.08	51.35	66.21
Replace GAT	w/o BERT _{base}	55.27	53.14	47.84	61.66
	w/ BERT _{base}	59.21	57.07	51.02	66.43

did not perform aggregation and update on the HDT, instead, the initial features were used to perform DocRE (−HDT). The results shown in Table 7 indicate the maximum performance loss caused by single component removal, with the HDT-BP experiencing reductions in the corresponding F1 and Ign-F1 scores of 2.16% and 2.22%, respectively, and HDT-BP_{BERT} experienced reductions of 1.61% and 1.58%. Finally, the model was trained by removing all three components simultaneously (−Both), and the HDT-BP/HDT-BP_{BERT} achieved only 52.78%/57.12% F1 scores and 50.54%/55.00% Ign-F1 scores on the DocRED development set, respectively. In summary, by utilizing either BERT or BiLSTM as the token embedding layer for the proposed model, the contribution trend of each component is consistent, which illustrates the overall effectiveness of the model from a bottom-up perspective.

In addition, to further verify that the performance of the proposed algorithm framework does not depend on a specific network operator (such as LSTM or GCN), we added two sets of operator replacement experiments in Table 7. These experiments involved replacing the LSTM and GCN operators in the algorithm framework described in Section 3 with GRU and GAT. The experimental results show that the replacement of LSTM with GRU negligibly affects the model performance. In the case of HDT-BP_{BERT}, replacing GCN with GAT caused a maximum F1 score loss of only 0.16%. Therefore, it can be concluded that substituting these operators has a trivial effect on the model performance. This phenomenon serves as evidence that the proposed model is relatively insensitive to the operator, and its performance primarily comes from the design of the algorithm framework.

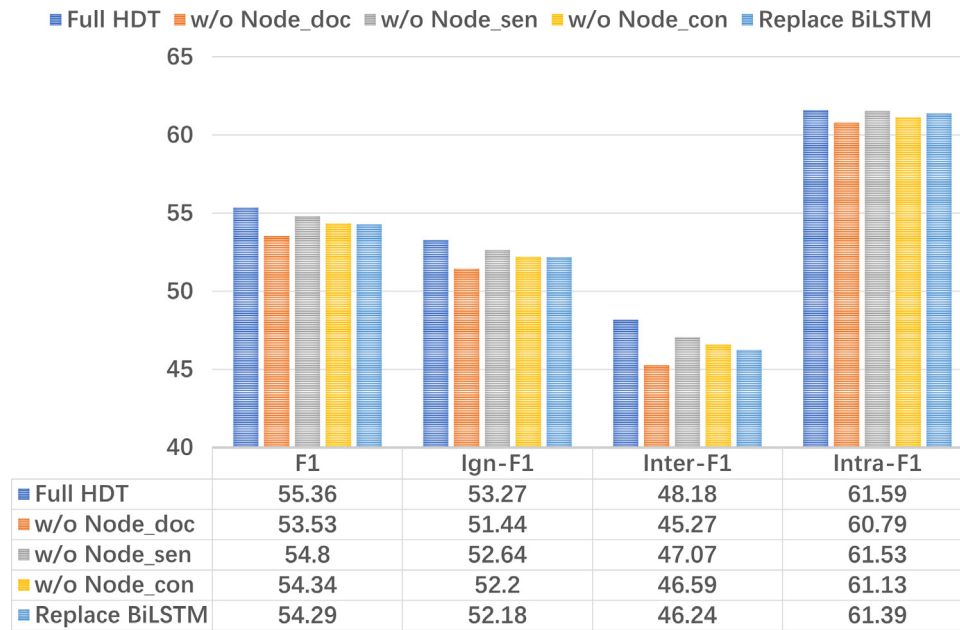


Fig. 7. Pruning study on the DocRED development set.

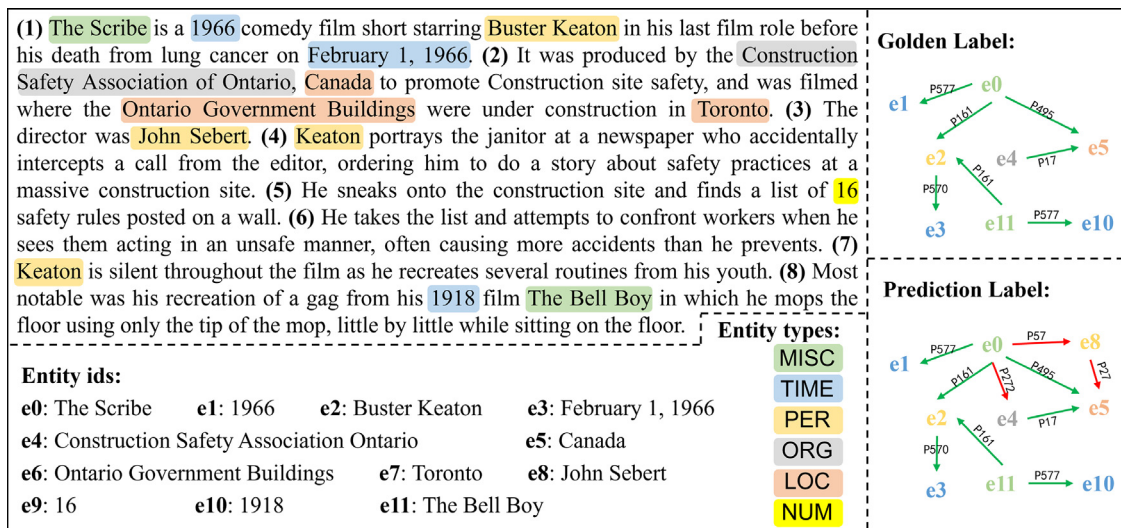


Fig. 8. Case analysis of the proposed model on DocRED. The model take the document as input and predicts the relations among different entities in different colors. The arrows in the label graph on the right represent the types and directions of the relations between entities.

5.7. Case analysis

To verify the performance of the proposed model, we conducted a visual case analysis by selecting a document sample completely at random from the development set. As shown in Fig. 8, the document consists of 8 sentences of different lengths, with the entities highlighted in different colors to distinguish their respective types. There are a total of 11 entities in the document belonging to 6 different types. These entities in the sample may be in the same sentence or distributed in different contexts. The graph on the right side of the figure shows the relationship information in the document, which includes both intra-sentence and inter-sentence relations and some one-to-many relations, posing a challenge for the RE model. The Golden Label in the graph represents the true relations between the entities, while the Prediction Label represents the predicted results of the proposed model, with the arrows indicating the relationship categories. The green arrows in the graph indicate

the correctly predicted relations, while the red arrows indicate the incorrect predictions. It can be seen from the analysis results that the proposed model extracts a total of 10 relations, among which 7 are predicted correctly with relatively good accuracy. Using entity e0, which has the most relations, as an example, all of the relationships from e0 to e1, e2, and e5 are accurately extracted. Despite the excellent performance of the model, there are still some inevitable mispredictions (e.g., e0 to e8), indicating that DocRE is still a very challenging task.

6. Conclusion

This paper introduces a new model for document-level relation extraction (DocRE) called the HDT-BP, which leverages the power of the Hierarchical Dependency Tree and Bridge Path. Unlike the conventional approach of modeling the entire document as a coarse-grained embedding, the proposed model takes a more nuanced approach by processing each sentence individually and

performing word embedding. To better represent the intricate structure dependency of a document, we introduce a fine-grained document reconstruction method that generates a sophisticated hierarchical dependency tree (HDT) containing four hierarchies of node features and three types of edge information. The HDT assigns independent weights to the contextual features of different hierarchies and regions, which participate in the deep reasoning of the model with the aggregation and update of the HDT. During the RE stage, the model builds an implicit bridge path in the document for entity pairs based on the relation bridge entity. The experimental results demonstrate that the proposed model achieves a competitive F1 score on the DocRED dataset and performs better on new relational facts that have not appeared in the training set compared to other models. Pruning and ablation studies further illustrate the importance of hierarchical features for DocRE. As we look towards the future, we envision expanding this framework to address the even more challenging task of dialog relation extraction.

CRedit authorship contribution statement

Qian Wan: Writing – original draft, Software, Methodology, Conceptualization. **Shangheng Du:** Data curation, Validation. **Yaqi Liu:** Formal analysis. **Jing Fang:** Investigation. **Luona Wei:** Writing – review & editing. **Sannyuya Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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